Matrices, Vectors & Complex Numbers

ANSWER KEY

Section 1: Operations with Complex Numbers

1 What is the product of the complex numbers $(4 - 3i)$ and $(2 + 5i)$, where $i^2 = -1$?

a. $23 + 14i$ ✓

b. $7 + 14i$

c. $23 - 14i$

d. $8 - 15i$

Expand: $4(2) + 4(5i) - 3i(2) - 15i^2 = 8 + 20i - 6i - 15(-1) = 8 + 14i + 15 = 23 + 14i$.

2 What is the simplified form of the quotient $\frac{5 - i}{2 + i}$?

a. $\frac{9 - 7i}{5}$ ✓

b. $\frac{11 - 7i}{5}$

c. $3 - i$

d. $\frac{9 + 3i}{5}$

Multiply numerator and denominator by conjugate $(2 - i)$: $\frac{(5 - i)(2 - i)}{(2 + i)(2 - i)} = \frac{10 - 5i - 2i + i^2}{4 - i^2} = \frac{10 - 7i - 1}{4 - (-1)} = \frac{9 - 7i}{5}$.

3 What is the value of i^{47}, where $i = \sqrt{-1}$?

a. 1

b. -1

c. i

d. $-i$ ✓

Divide power by 4: $47 = 4(11) + 3$. Thus $i^{47} = (i^4)^{11} \cdot i^3 = 1 \cdot (-i) = -i$.

Section 2: Modulus & Graphing in the Complex Plane

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4 What is the modulus (absolute value) of the complex number $z = -5 + 12i$?

a. 7

b. 13 ✓

c. 17

d. $\sqrt{119}$

Modulus $|z| = \sqrt{a^2 + b^2} = \sqrt{(-5)^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$.

5 What is the distance in the complex plane between $z_1 = 3 - 2i$ and $z_2 = -1 + 4i$?

a. $2\sqrt{13}$ ✓

b. $\sqrt{20}$

c. 10

d. 5

Distance
$= |z_1 - z_2| = |(3 - (-1)) + (-2 - 4)i| = |4 - 6i| = \sqrt{4^2 + (-6)^2} = \sqrt{16 + 36} = \sqrt{52} = 2\sqrt{13}$.

6 If $z = 2 + 2i$, what is the argument θ of z in radians, where $0 \le \theta < 2\pi$?

a. $\frac{\pi}{6}$

b. $\frac{\pi}{4}$ ✓

c. $\frac{\pi}{3}$

d. $\frac{3\pi}{4}$

Both real and imaginary parts are positive (Quadrant I). $\tan\theta = \frac{b}{a} = \frac{2}{2} = 1 \Rightarrow \theta = \frac{\pi}{4}$.

Section 3: Matrix Operations & Determinants

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7 Given matrices $A = [[3, -1], [2, 4]]$ and $B = [[1, 5], [-2, 3]]$, what is the matrix $2A - B$?

a. $[[5, -7], [6, 5]]$ ✓

b. $[[5, 3], [2, 5]]$

c. $[[7, -7], [6, 11]]$

d. $[[5, -7], [2, 5]]$

$2A = [[6, -2], [4, 8]]$. Subtract B: $[[6-1, -2-5], [4-(-2), 8-3]] = [[5, -7], [6, 5]]$.

8 What is the determinant of the 2×2 matrix $M = [[4, 6], [-2, 3]]$?

a. 0

b. 12

c. 24 ✓

d. 36

Determinant $\det(M) = ad - bc = 4(3) - (6)(-2) = 12 - (-12) = 24$.

9 For what value of k will the matrix $K = [[k, 6], [2, 3]]$ NOT have an inverse?

a. 1

b. 2

c. 3

d. 4 ✓

A matrix is singular (no inverse) when its determinant is zero:
$k(3) - (6)(2) = 0 \Rightarrow 3k - 12 = 0 \Rightarrow 3k = 12 \Rightarrow k = 4$.

Section 4: Vector Operations & Dot Product

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10 Vectors $\mathbf{u} = \langle 3, -4 \rangle$ and $\mathbf{v} = \langle -2, 5 \rangle$ are given. What are the components of $2\mathbf{u} + \mathbf{v}$?

a. $\langle 4, -3 \rangle$ ✓

b. $\langle 1, 1 \rangle$

c. $\langle 4, 3 \rangle$

d. $\langle -1, -3 \rangle$

$2\mathbf{u} = \langle 6, -8 \rangle$. Adding $\mathbf{v}$: $\langle 6 + (-2), -8 + 5 \rangle = \langle 4, -3 \rangle$.

11 What is the dot product of the vectors $\mathbf{a} = \langle 5, -2 \rangle$ and $\mathbf{b} = \langle 4, 7 \rangle$?

a. 6 ✓

b. 26

c. -6

d. 34

$\mathbf{a} \cdot \mathbf{b} = (5)(4) + (-2)(7) = 20 - 14 = 6$.

12 What is the magnitude of the vector $\mathbf{w} = \langle -9, 12 \rangle$?

a. 13

b. 15 ✓

c. 17

d. 21

Magnitude $\|\mathbf{w}\| = \sqrt{(-9)^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15$.

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